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Color image encryption by using Arnold transform and color-blend operation in discrete cosine transform domains

Zhengjun Liu [a,*], Lie Xu [a], Ting Liu [a], Hang Chen [a], Pengfei Li [a], Chuang Lin [c], Shutian Liu [b]

[a] Department of Automation Measurement and Control Engineering, Harbin Institute of Technology, Harbin 150001 PR China
[b] Department of Physics, Harbin Institute of Technology, Harbin 150001 PR China
[c] Department of Network Engineering, School of Software, Dalian University of Technology, Dalian 116620 PR China

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ABSTRACT

A color image encryption algorithm is designed by use of Arnold transform and discrete cosine transform (DCT). The RGB components of the color image are scrambled by Arnold transform at the aspect of pixel sequence. The scrambled RGB components are exchanged and mixed randomly under the control of a matrix defined by random angle. DCT is employed for changing the pixel values of color image. In this encryption scheme the operations mentioned above are performed twice continuously. The parameters of Arnold transform and the random angle serve as the key of the color image encryption method. Some numerical simulations are made to test the validity and capability of the color encryption algorithm.

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1. Introduction

Hiding secret data or information becomes the main goal for image encryption and image watermarking. A random reversible process is the kernel designing information security algorithm. Here a random operation can be classified into two kinds of formats, namely randomizing the value and sequencing of data (or image). In double random phase encoding (DRPE) technology [1–8], the data of the original image is modulated and changed by random phase function, which is regarded as the operation randomizing value of secret information. Moreover some optical processes or transforms [9–23] have been considered for changing the pixel value of image. The scrambling operations, such as Arnold transform and jigsaw transform, are used to make the change of sequence of image pixels [24–33]. A secret image can be randomized by an encryption scheme composed of random reversible process.

Recently some color image encryption algorithms [33–39] have been reported and researched. In some encryption algorithms, color image is decomposed into the components of red, green and blue (RGB). The corresponding color components are encoded by an encryption method of gray-level image. The RGB components of color image are scrambled and encoded by Arnold transform and DRPE [33], respectively. A secure information display system of multi-color visual cryptography [34] has been reported. Two encryption methods of color image have been proposed by use of the DRPE composed of fractional Fourier transform [35,36]. Color image is considered as three homophonous images and is imported into optical encryption system [37,38]. A vector model composed of the RGB components of color has been employed for designing the encryption algorithm [39].

In this paper, we propose a color image encryption algorithm by using the Arnold transform and discrete cosine transform. An original color image is blocked and is scrambled by Arnold transform. A 3×3 matrix defined by a random angle is used for exchanging the pixel value of RGB components of color image. Subsequently the scrambled and exchanged image is transformed by DCT. Arnold transform, exchanging pixel and DCT are performed twice in this encryption method. The data of random angle is the main key of encryption. The parameters of Arnold transform are the additional key for increasing security. Numerical simulation is performed for validating the performance of the proposed color image encryption.

The rest of this paper is organized in the following sequence. In Section 2, the proposed encryption and decryption algorithms of color image are represented in detail. In Section 3, numerical simulation is given to test the validity and security of this encryption scheme. Concluding remarks are summarized in the final section.

2. Encryption algorithm

In the design of this color image encryption, both scrambling sequence and changing pixel value are considered and utilized. Fig. 1 gives the flowchart of this color image encryption scheme. Firstly Arnold transform is introduced for scrambling the pixel position

* Corresponding author. Tel.: +86 45186413151-8008; fax: +86 45186415146.
E-mail addresses: zjliuv@gmail.com, zjliu@hit.edu.cn (Z. Liu).

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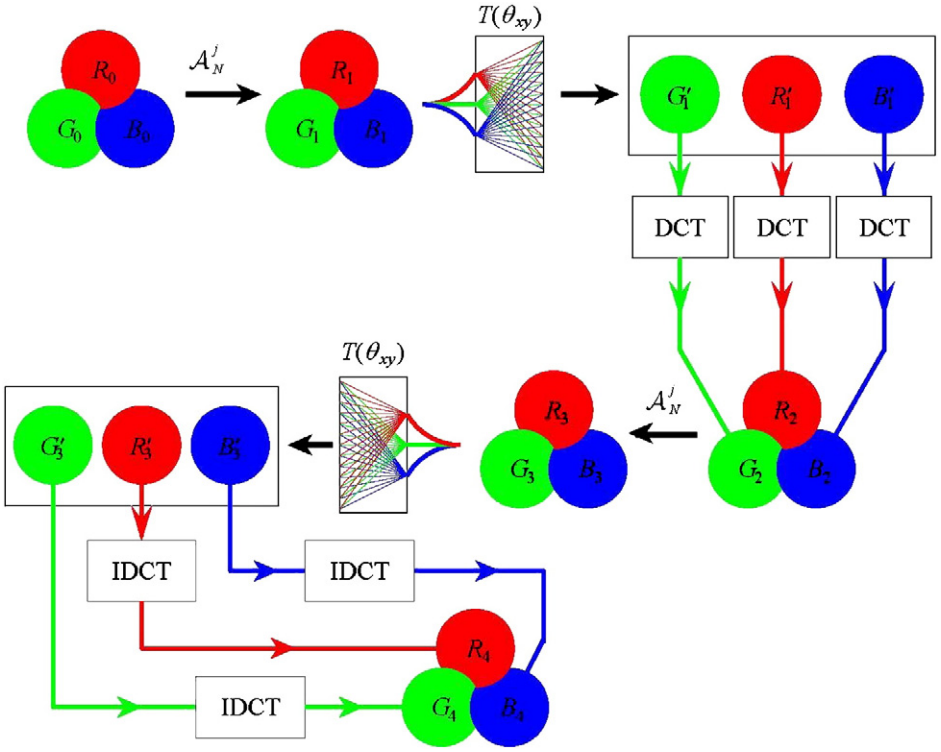


Fig. 1. The flowchart of the color image encryption algorithm.

of the RGB components of color image. The Arnold transform $\mathcal{A}_N^j$ is defined as [40]

$$\begin{bmatrix} m_{j+1} \\ n_{j+1} \end{bmatrix} = \mathrm{mod}\left(\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} m_j \\ n_j \end{bmatrix}, N \right), \quad j = 1, 2, 3, \ldots \tag{1}$$

where $[m_j, n_j]^t$ and $[m_{j+1}, n_{j+1}]^t$ represent the position vector of image pixel shifted before and after, respectively. The parameter N is the size of imported target image. The period of the transform [40] depends on the parameter N. Before Arnold transform is executed, the original color image is blocked into several sub-images as Fig. 2. The performing times of Arnold transform are fixed at different values for enhancing the security of image encryption. Other blocking scheme can also be designed and employed for replacing the blocking scheme in Fig. 2. After the Arnold transform of the blocked sub-images is executed, the three color components of the

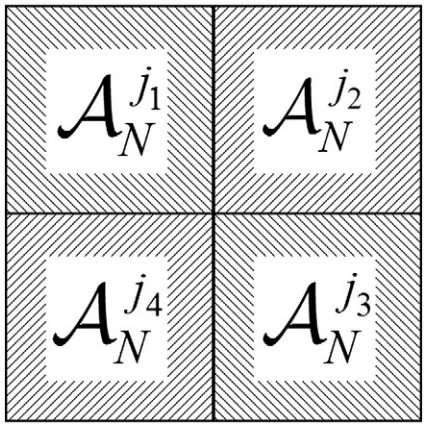


Fig. 2. The block scheme of the original image.

original image ($R_0(x,y)$,$G_0(x,y)$ and $B_0(x,y)$) become the functions ($R_1(x,y)$,$G_1(x,y)$ and $B_1(x,y)$). Thereby, the pixel sequence of the original image is randomized at a local area.

Following Arnold transform, an affine transform is introduced and defined with an angle θ as

$$\begin{aligned} T(\theta) &= \begin{bmatrix} t_{11} & t_{12} & t_{13} \\ t_{21} & t_{22} & t_{23} \\ t_{31} & t_{32} & t_{33} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sin\theta & -\cos\theta \\ 0 & \cos\theta & \sin\theta \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta & \sin^2\theta & -\sin\theta\cos\theta \\ -\sin\theta & \sin\theta\cos\theta & -\cos^2\theta \\ 0 & \cos\theta & \sin\theta \end{bmatrix}, \end{aligned} \tag{2}$$

where the function $T(\theta)$ is an orthogonal matrix. In fact other orthogonal 3×3 matrices can be introduced for replacing the matrix $T(\theta)$ in Eq. (2). The inverse matrix of $T(\theta)$ can be calculated as

$$\begin{aligned} T^{-1}(\theta) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sin\theta & \cos\theta \\ 0 & -\cos\theta & \sin\theta \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin^2\theta & \sin\theta\cos\theta & \cos\theta \\ -\sin\theta\cos\theta & -\cos^2\theta & \sin\theta \end{bmatrix}. \end{aligned} \tag{3}$$

The functions ($R_1(x,y)$,$G_1(x,y)$ and $B_1(x,y)$) are re-combined by the matrix $T(\theta)$ and $T^{-1}(\theta)$ as

$$\begin{bmatrix} R_1'(x,y) \\ G_1'(x,y) \\ B_1'(x,y) \end{bmatrix} = T\left(\theta_{xy}\right) \begin{bmatrix} R_1(x,y) \\ G_1(x,y) \\ B_1(x,y) \end{bmatrix} \tag{4a}$$

and

$$\begin{bmatrix} R_1(x,y) \\ G_1(x,y) \\ B_1(x,y) \end{bmatrix} = T^{-1}\left(\theta_{xy}\right) \begin{bmatrix} R_1'(x,y) \\ G_1'(x,y) \\ B_1'(x,y) \end{bmatrix}, \tag{4b}$$

where the symbol θ_{xy} is a random angle function depending on the variables x and y. Eqs. (4a) and (4b) are used for encrypting and decrypting image in this paper. Eqs. (4a) and (b) can exchange the pixel values of color image. Meantime the random angle function θ_{xy} is introduced and regarded as the key of encryption algorithm via Eqs. (4a) and (4b).

DCT is employed in the process of image encryption. After the Arnold transform of blocked sub-image is performed, the bound between different blocking images will reveal the blocking scheme. DCT can change the distribution of pixel value of whole image for obtaining a random output pattern. Another reason using DCT is that this transform is defined in real number field. Thus output data can be encoded with real number in this encryption algorithm. When the DCT of the three color components is achieved, the functions R_2, G_2 and B_2 are received.

In the following steps, the functions R_3, G_3 and B_3 are calculated by blocking operation and Arnold transform, in which blocking scheme is fixed at the same with Fig. 2. Here the parameters of Arnold transform are given as Fig. 2. Subsequently the functions R'_3, G'_3 and B'_3 are obtained by using Eq. (4a). The inverse DCT is utilized for getting the color components of the final encrypted image (R_4, G_4 and B_4). The image decryption process can be designed according to the reverse direction of image encryption flowchart shown in Fig. 1. The inverse transforms and operations will be used for recovering original secret image.

The computing complexity of the proposed encryption method depends on the Arnold transform, DCT and the color-blend defined in Eqs. (4a) and (4b). The two transforms can be implemented with some published codes written by computing language (such as MATLAB or C language). For the color-blend operation, Eqs. (4a) and (4b) can be re-expressed as a set of equations, in which every equation indicates a two-dimensional function relation, for the design of calculation procedure.

Compared to classical DRPE, the random angle θ_{xy} serves as the main key and re-combines the pixel values of color components of color image in this encryption method. Moreover the parameters of Arnold transform are regarded as the additional key for enhancing the security of encryption algorithm. In the DRPE method, random phase function is attached into input image and its transformed spectrum (here the transform is taken at anyone between Fourier transform, fractional Fourier transform and other optical transforms). Under the control of random angle θ_{xy} and random phase function, the pixel values of image are changed randomly.



Fig. 3. The original color image used in numerical simulation.

3. Numerical simulation

A color Lena picture having 256×256 pixels is regarded as an original image and shown in Fig. 3. The random data of angle function θ_{xy} satisfies uniform distribution in the range $[0,2\pi]$. The parameters of Arnold transform $\{j_1,j_2,j_3,j_4\}$ are taken at {32,55,63,43}. When $N=128$, the period of Arnold transform A^j_{128} equals 96. The corresponding encrypted image is displayed in Fig. 4(a). By use of correct keys (θ_{xy} and $\{j_1,j_2,j_3,j_4\}$), the decrypted image is represented in Fig. 4(b), which is the same with the original image.

To calculate the difference between the decrypted image I_d and original image I_o, the peak signal-to-noise ratio (PSNR) function is introduced and expressed as

$$\begin{aligned} \mathrm{PSNR} &= \mathrm{psnr}(I_d, I_o) \\ &= 10\log_{10}\frac{255^2 M\times K}{\sum_{\forall m,k}[I_d(m,k)-I_o(m,k)]^2}\ (\mathrm{dB}), \end{aligned} \tag{5}$$

where M and K give the size of the two images. To indicate the quality of decrypted result in vision, the recovered image obtained with the boundary value of parameter is also displayed to assist PSNR.

The parameters (j_1,j_2,j_3 and j_4) are regarded as the additional key of this encryption scheme. As an example, the various values of the parameter j_1 in a period are tested in a decryption process. The corresponding PSNR curve is shown in Fig. 5. When $j_1=33$,which is

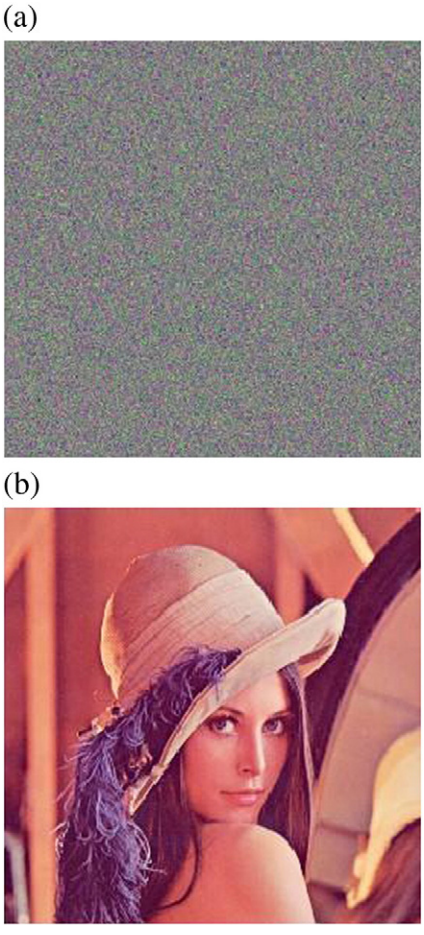


Fig. 4. The image encryption and decryption results: (a) encrypted image, (b) decrypted image.

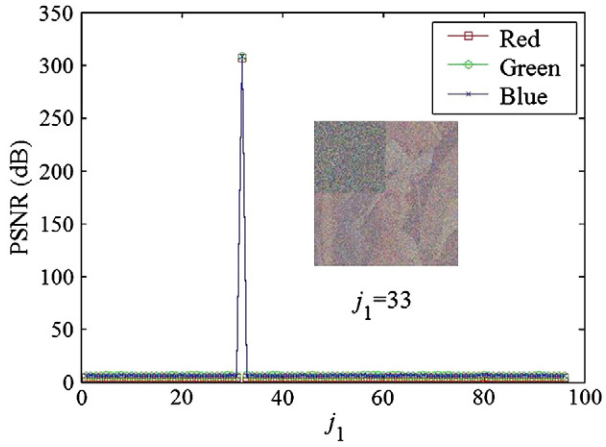


Fig. 5. The PSNR calculated by the different values of the parameter j_1 of Arnold transform.

closed to the correct value, the decrypted image is given in Fig. 5. The left-top part of the image is random pattern and other area has some noise. From Fig. 5, the parameter $j_1$regarding as the additional key can protect the corresponding sub-image blocked.

When a part of the data of the angle θ_{xy} is known by an illegal user, the decryption in this case is considered and simulated. Moreover, we suppose that the parameters of Arnold transform and decryption process are learned by the attacker. Two examples are tested with a part-correct angle θ_{xy}. The corresponding results are illustrated in

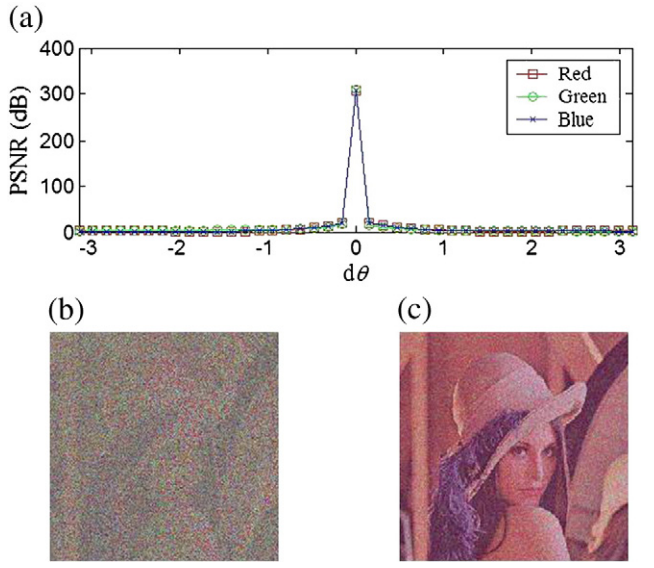


Fig. 7. The decryption result calculated with various values of dθ: (a) PSNR curve, (b) dθ = 1, and (c) dθ = π/20.

Fig. 6, in which the data at the right side and upper triangular part are determined. From Fig. 6, the outline of the original secret information at the area, in which the value of the angle θ_{xy} is known, appears hazily

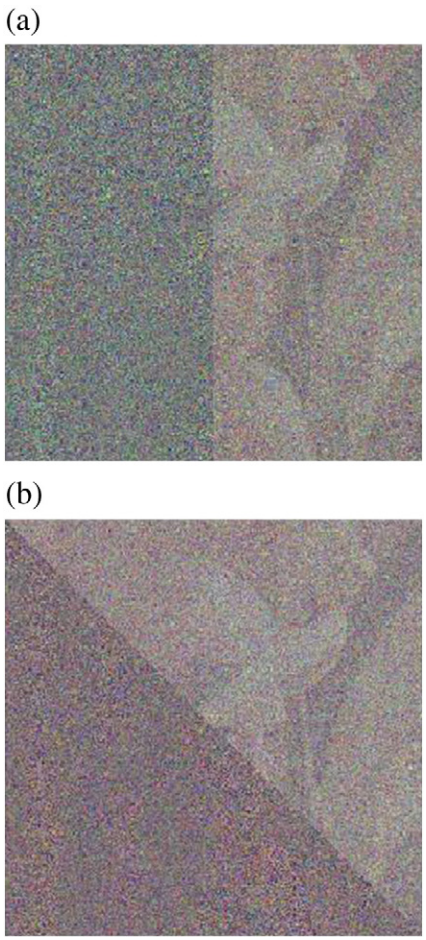


Fig. 6. The decryption with a half of correct data of the angle θ_{xy}: (a) the data at the left side is wrong, (b) the data at the lower triangular part is wrong.

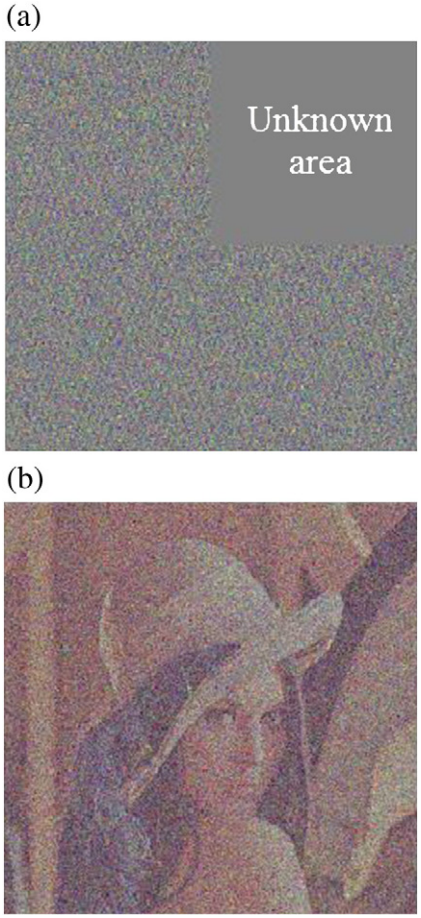


Fig. 8. The test of occlusion attack: (a) the occluded encrypted image and (b) the recovered image. Here the PSNR values of three color components (red, green and blue) are equal to 10.38 dB, 11.58 dB and 11.61 dB, respectively.

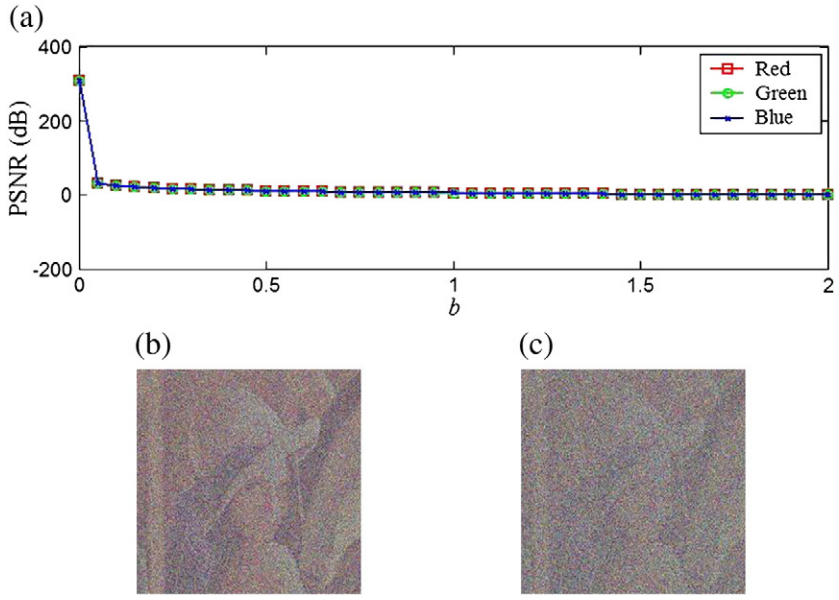


Fig. 9. The test of noise attack: (a) PSNR curve, (b) decrypted image obtained with $b=1$, and (c) decrypted image obtained with $b=2$.

and has random color information. At the range composed of unknown values of angle θ_{xy}, the decrypted pattern is random. This phenomenon is similar to the result reported in [32].

Now we pay attention to a worst case for the angle θ_{xy}, namely the data of the angle θ_{xy} used in image decryption has a constant regarding as the error comparing the correct value. Here the decryption process and the parameters of Arnold transform are right in calculation. The changed angle θ'_{xy} can be represented as follows

$$\theta'_{xy} = \theta_{xy} + d\theta, \tag{6}$$

where the parameter $d\theta$ denotes the error between the right and wrong angle functions. To our best knowledge, the original image can be decrypted completely, if the second phase in DRPE [1] is defined according to Eq. (6). Because the period of the function $T(\theta_{xy})$ is equal to 2π, the interval $[-\pi,\pi]$ is adopted and serves as the sampling range of the parameter $d\theta$. Here the sampling step length is taken at $\pi/20$. Recovering the secret image by use of the angle θ'_{xy}, the corresponding results are given in Fig. 7(a). When $d\theta=1$, the decrypted image in Fig. 7(b) has the main outline of secret image and random color. Another decryption result in Fig. 7(c) is closed to the original image. From Fig. 7, the color information of the original image can be decrypted in a small interval.

The robustness of this encryption will be checked by occlusion attack and noise attack. The decryption process is performed with a correct key from an encrypted image occluded partly, which is shown in Fig. 8(a). Here the values of the occluded pixels at the right-top corner are replaced with 0 in simulation. The decrypted image is represented in Fig. 8(b), from the main information of the original image can be recognized in vision.

The model of noise attack is simulated and defined as follows

$$I'(x,y) = I(x,y)\left[1 + b\cdot\sigma_{0,1}(x,y)\right], \tag{7}$$

where $I(x,y)$ and $I'(x,y)$ represent the color component image (red, green or blue) of the encrypted data before and after adding noise, respectively. Parameter b is the coefficient controlling the noise intensity. Function $\sigma_{0,1}(x,y)$ is a random data satisfying uniform distribution. The mean value and standard deviation of $\sigma_{0,1}(x,y)$ equal 0 and 1, respectively. By using the images $I'(x,y)$ generated with various values of parameter b, the PSNR curve is computed and drawn in Fig. 9(a). When $b=1$ and $b=2$, the two decrypted images are illustrated in Fig. 9(b) and (c), in which the outline of the secret image can be identified.

4. Conclusion

We have represented a kind of color image encryption algorithm by use of Arnold transform and color-blend operation in discrete cosine transform domains. The Arnold transform scrambles the pixel sequence of color image for several sub-images at local area. The color-blend operation defined by a 3×3 matrix combines the three color components of color image again. The discrete cosine transform is employed for changing the pixel value of color image at whole spatial distribution. The encryption process includes two sets of the operations mentioned above. The angle data in the color-blend operation is regarded as the main key of this algorithm. The parameters of Arnold transform serve as the additional key. Some numerical simulations have demonstrated the security and robustness of the proposed encryption scheme.

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